

Notes: Aiello (JFE, 2022)

Financially Constrained Mortgage Servicers

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1 Big picture

- **Question.** What happens when mortgage servicers are financially constrained while managing delinquent securitized mortgages on behalf of MBS investors?
- **Main mechanism.** Mortgage servicers must advance missed monthly principal and interest payments to MBS investors when borrowers become delinquent. These servicing advances are recovered when the servicer completes a loss-mitigation action, such as foreclosure or modification. Therefore, financially constrained servicers have strong incentives to act quickly to stop making advances.
- **Agency problem.** The servicer decides whether and when to modify or foreclose, but the investor bears the ultimate loan cash-flow consequences. Advances reduce some agency conflicts by making servicers bear short-run costs, but they also make servicer balance sheets important.
- **Core empirical finding.** A one-standard-deviation increase in servicer financial constrainedness substantially raises the probability that a defaulted loan receives an intervention. In the main IV specification, the intervention probability rises by about 14 percentage points.
- **Foreclosure and modification.** The effect operates through both foreclosure and modification. Financially constrained servicers are about 9 percentage points more likely to foreclose and about 5 percentage points more likely to modify after the first delinquency episode.
- **Investor value.** The aggressive interventions are not investor-value enhancing. The paper estimates that financial constraints reduce investor NPV by roughly \$21,663 per defaulted loan in the main IV specification.
- **Economic magnitude.** The paper argues that servicer financial constraints account for about 20% of the total investor value destroyed per defaulted loan during the financial crisis.
- **Agency-friction heterogeneity.** The effect of constraints is much weaker for high-quality delinquent loans, where the servicer's optimal action is easier for investors to monitor and agency frictions are lower.
- **Capacity constraints.** Operational capacity also matters. A servicer facing high capacity constraints is less able to respond to financial constraints by rapidly modifying or foreclosing loans.
- **Takeaway.** Mortgage servicing is not a passive administrative function. Servicer financial condition can distort default management, destroy investor value, and affect foreclosure outcomes in housing crises.

2 Data and measurement

2.1 Mortgage servicing and servicing advances

- In a securitized mortgage, the investor owns the cash-flow claim, but a mortgage servicer manages billing, collection, escrow accounts, delinquency resolution, modifications, and foreclosure.
- When a borrower misses payments, the servicer typically advances scheduled principal and interest to the MBS investor.
- Advances are reimbursed when the loan is modified, foreclosed, liquidated, paid off, repurchased, or otherwise resolved.
- The advance obligation gives the borrower time to self-cure or pay off, but it also creates a financing burden for the servicer.

Interpretation: The servicer effectively provides short-term financing to delinquent borrowers. If the servicer is financially constrained, it has an incentive to shorten the delinquency episode even if waiting would preserve investor value.

2.2 Loan-level performance data

- The paper uses a rich loan-level dataset covering approximately 90% of mortgage loans sold into non-agency RMBS transactions before the crisis.
- Non-agency RMBS loans are central because they account for a large share of crisis-era delinquencies, modifications, and foreclosures.
- The data track monthly loan performance, delinquency status, modifications, foreclosures, liquidation, bankruptcy, repurchases, and other outcomes.
- The main sample conditions on loans that defaulted and studies the outcome of the borrower’s first episode of delinquency.

Interpretation: Conditioning on the first delinquency episode avoids contamination from prior servicer interventions and focuses on the first major decision opportunity after default.

2.3 Outcome definitions

The paper distinguishes several outcomes after a borrower’s first delinquency episode:

- **Self cure:** the borrower becomes current without a modification, pay-off, foreclosure, or liquidation.
- **Paid-in-full:** the loan is paid off, typically through sale or refinancing.
- **Modified:** the loan terms are modified.
- **Foreclosed:** the servicer initiates or completes foreclosure-related action.
- **Liquidation:** the collateral is liquidated through charge-off or short sale.
- **Bankruptcy or repurchase:** alternative resolution categories.

Interpretation: Self-cure and pay-off are generally outcomes where the borrower resolves the delinquency without costly servicer intervention. Modification and foreclosure are the key actions through which a constrained servicer can reduce advance obligations.

2.4 Financial constrainedness measure

The main measure of servicer financial constrainedness is:

$$\Delta AdvFrac_{s,t} = \Delta \left(\frac{Advances_{s,t}}{ServicerPortfolioBalance_{s,t}} \right), \quad (1)$$

where s indexes the servicer and t is measured around the borrower’s first delinquency.

- *Advances* are outstanding servicing advances.
- *ServicerPortfolioBalance* is the outstanding balance of the servicer’s portfolio.
- The measure is standardized in the regression samples.

Interpretation: A rising advances-to-portfolio ratio means that the servicer is financing more delinquent payments relative to the size of its mortgage servicing book. This is interpreted as higher financial constrainedness.

2.5 Instrument: servicer portfolio housing price returns

Because advances are endogenous to servicer decisions, the paper instruments financial constrainedness using housing price declines in the servicer’s broader portfolio, excluding the focal loan’s local housing market:

$$Z_{i,s,t} = \text{ServicerPortfolioHPR}_{s,t}^{-CBSA(i)}. \quad (2)$$

- A negative housing price shock in the servicer’s other markets increases delinquencies and advance obligations elsewhere in the portfolio.
- Excluding the focal loan’s CBSA removes the direct local housing shock affecting the focal borrower.
- The instrument is also implemented with servicer-level specifications and alternative measures such as CDS spreads.

Interpretation: The logic is that housing price declines outside the focal market tighten the servicer’s balance sheet but should not directly determine the focal borrower’s outcome, conditional on controls and fixed effects.

2.6 Controls and fixed effects

The empirical specifications include a rich set of controls:

- loan and borrower characteristics such as FICO, LTV, ARM status, owner occupancy, purchases, single-family status, and documentation status;
- local housing price returns;
- servicer fixed effects;
- zip-year-credit-category fixed effects;
- year fixed effects in some specifications;
- delinquency-count controls capturing servicer-level capacity pressure.

Interpretation: These controls aim to separate servicer financial constraints from local borrower quality, loan risk, housing market conditions, and servicer-specific business models.

2.7 Investor NPV

The paper constructs a loan-level investor NPV using observed monthly cash flows:

$$NPV_i = \sum_{m \geq 0} \frac{CF_{i,m}}{(1 + r_i)^m}, \quad (3)$$

where $CF_{i,m}$ are investor cash flows from loan i and r_i is a discount rate tied to the loan contract or servicing-strip-adjusted rate.

Interpretation: Investor NPV is the most direct welfare object for MBS investors. It asks whether the servicer’s actions improve or destroy the present value of the investor’s claim.

3 Empirical specifications

3.1 First-stage regression

The first stage relates servicer financial constrainedness to the excluded housing-price instrument:

$$\Delta AdvFrac_{s(i),t} = \pi Z_{i,s,t} + \Gamma X_{i,t} + \Phi_s + \Lambda_{z,t,c} + u_{i,t}. \quad (4)$$

- $Z_{i,s,t}$ is the servicer portfolio housing price return excluding the focal loan's CBSA.
- $X_{i,t}$ are loan and borrower controls.
- Φ_s are servicer fixed effects.
- $\Lambda_{z,t,c}$ are zip-year-credit-category fixed effects.

Interpretation: The first stage tests whether exogenous housing price deterioration in a servicer's other markets increases the servicer's advance burden.

3.2 Baseline second-stage outcome equation

For binary outcomes such as intervention, foreclosure, modification, paid-in-full, or self-cure, the second stage is:

$$Y_{i,t} = \alpha + \beta \widehat{\Delta AdvFrac}_{s(i),t} + \Gamma X_{i,t} + \Phi_s + \Lambda_{z,t,c} + \varepsilon_{i,t}. \quad (5)$$

- $Y_{i,t}$ is an outcome after the first delinquency episode.
- β measures the causal effect of servicer financial constrainedness.
- The paper implements this as a linear probability model with two-stage least squares.

Interpretation: A positive β for foreclosure or modification means constrained servicers intervene more aggressively. A negative β for self-cure or paid-in-full means borrowers are less likely to resolve delinquency without costly intervention.

3.3 Ultimate outcomes

The paper also studies ultimate outcomes over the life of the loan:

$$UltimateY_i = \alpha + \beta \widehat{\Delta AdvFrac}_{s(i),t} + \Gamma X_{i,t} + \Phi_s + \Lambda_{z,t,c} + \varepsilon_i. \quad (6)$$

Interpretation: First-episode actions may be reversed or followed by later events. Ultimate outcomes test whether financial constraints have durable effects beyond the initial intervention window.

3.4 Agency-friction heterogeneity

The paper defines high-quality loans as loans with relatively strong borrower fundamentals, such as high FICO and low LTV. It estimates:

$$Y_i = \alpha + \beta_1 \widehat{\Delta AdvFrac}_{s(i),t} + \beta_2 HighQuality_i + \beta_3 \left(\widehat{\Delta AdvFrac}_{s(i),t} \times HighQuality_i \right) + \Gamma X_i + FE + \varepsilon_i. \quad (7)$$

Interpretation: High-quality loans have lower agency frictions because monitoring costs are lower and the investor can more easily infer what a prudent servicer should do. The paper predicts $\beta_3 < 0$ for intervention and foreclosure.

3.5 Investor-value regression

Investor value is estimated using:

$$NPV_i = \alpha + \beta \widehat{\Delta AdvFrac}_{s(i),t} + \Gamma X_i + FE + \varepsilon_i. \quad (8)$$

The high-quality heterogeneity version adds the same interaction structure:

$$NPV_i = \alpha + \beta_1 \widehat{\Delta AdvFrac}_{s(i),t} + \beta_2 HighQuality_i + \beta_3 \left(\widehat{\Delta AdvFrac}_{s(i),t} \times HighQuality_i \right) + \Gamma X_i + FE + \varepsilon_i. \quad (9)$$

Interpretation: This regression asks whether interventions caused by servicer constraints actually preserve or destroy investor value.

3.6 Conditional performance of foreclosures and modifications

The paper also conditions on specific actions:

- foreclosure liquidation loss severity;
- modification interest rate reduction;
- 3-month and 12-month redefault rates;
- investor NPV conditional on foreclosure, modification, or liquidation.

Interpretation: These tests evaluate whether constrained servicers simply do more interventions or also perform worse conditional on acting.

3.7 Capacity-constraint specification

Capacity constraints are proxied by changes in the delinquency count relative to the servicer’s portfolio:

$$CapacityPressure_{s,t} = \Delta \left(\frac{DelinquentLoans_{s,t}}{ServicerPortfolioLoanCount_{s,t}} \right). \quad (10)$$

The paper estimates baseline effects separately for low- and high-capacity-constraint subsamples.

Interpretation: Financial constraints and operational constraints interact. A servicer may want to intervene faster but be unable to do so if staffing, processing, and legal capacity are already strained.

4 Identification and empirical design

4.1 Relevance of the instrument

- Falling housing prices in a servicer’s other markets increase delinquencies elsewhere in the servicer’s portfolio.
- More delinquencies elsewhere require more servicing advances.
- Table 2 shows a strong first-stage relation between servicer portfolio housing price returns and advance growth.

Interpretation: The first stage is economically intuitive: house-price declines create delinquency waves and therefore cash needs for servicers.

4.2 Exclusion restriction

The exclusion restriction is that housing price movements in the servicer’s other markets affect the focal loan outcome only through the servicer’s financial constraint, not directly through borrower quality or local house prices.

- The focal loan’s CBSA is excluded from the instrument.
- Zip-year-credit-category fixed effects absorb local market shocks.
- Loan and borrower controls absorb observable credit quality.
- Servicer fixed effects absorb persistent differences in servicer business models.

Interpretation: The paper’s design is strongest if loans in the same local market are comparable but are serviced by entities with different exposure to housing price shocks elsewhere.

4.3 Why agency-friction tests matter

- A servicer’s financial constraint should matter more when the investor cannot easily monitor the optimal action.
- High-quality delinquent loans are easier cases: borrowers have better ability to self-cure or pay off.
- The paper finds smaller financial-constraint effects for high-quality loans.

Interpretation: This heterogeneity connects the empirical results to the agency-conflict mechanism rather than merely documenting more foreclosures during bad servicer states.

4.4 Unobservable borrower quality tests

The paper tests whether the instrument is correlated with borrower quality at origination using:

- one-year and five-year delinquency probabilities;
- months until first delinquency;
- early payment behavior in months 1–6, 4–6, and 4–9.

Interpretation: Table 9 shows no strong systematic relation between instrumented servicer constraints and these borrower-quality proxies, reducing concern that the instrument is picking up hidden borrower risk.

4.5 Capacity constraints as an additional mechanism

- High capacity pressure can limit how quickly servicers can process modifications or foreclosures.
- If financial constraints increase the desire to intervene but capacity constraints limit ability, the financial-constraint effect should be stronger when capacity pressure is low.
- Table 10 confirms this pattern.

Interpretation: This result refines the main mechanism: balance-sheet pressure creates incentives, but operational capacity determines how strongly those incentives translate into actions.

4.6 Limitations

- Financial constrainedness is measured using servicing advances, which are partly endogenous; the IV strategy is therefore central.
- The sample focuses on non-agency RMBS, where servicing frictions may be especially severe.
- Investor NPV is constructed from observed cash flows and modeling choices about discount rates and terminal values.
- Servicer capacity is difficult to measure directly, so delinquency-count pressure is a proxy.
- The analysis is most informative for crisis-era default management, not necessarily normal housing markets.

Interpretation: The paper’s strength is loan-level granularity and a targeted instrument. The main caveat is that all empirical measures of servicer constraint and servicer capacity are proxies.

5 Tables and figures

The paper's evidence is concentrated in tables. The tables below are directly cropped from the paper and grouped compactly.

5.1 Table 1: Summary statistics

Read:

- Panel A shows that defaulted loans have average original balances around \$249,000 and median original balances around \$189,000.
- Defaulted loans have average FICO scores near 668 and average LTV around 72.
- Panel B shows that 78.2% of loans in the sample default. Among defaulted loans, 26.3% self-cure, 39.6% are paid in full, 7.1% are modified, and 24.0% are foreclosed.
- Average investor NPV is much lower for foreclosed loans than paid-in-full loans: approximately \$122,774 versus \$251,460.
- Panel C shows that different outcomes are associated with different ex ante loan characteristics.

Economic message: Default outcomes differ dramatically in value. This motivates studying whether servicer incentives distort the path that delinquent loans take.

Table 1
Summary statistics.
This table reports summary statistics relating to the loan population utilized in this paper. Column (1) of Panels A and B report results relating to the population of BlackBox loans with valid covariate and servicer values. The remainder of the columns relate to the loans that experienced at least one qualifying delinquent episode. Panel A reports summary statistic data. Panel B reports performance statistics at the loss mitigation outcome level associated with the first delinquent episode. The average months to completion are the number of months between first default and the month of the borrower's third consecutive current payment, the pay off date, modification effective date, foreclosure sale date, liquidation date, bankruptcy petition filing date, or the date of repurchase as applicable. Panel C replicates Panel A statistics grouping by loss mitigation outcomes.

Panel A: summary of loan characteristics						
	BlackBox Mean (1)	Average (2)	Median (3)	Sample Std. Dev. (4)	10th% (5)	90th% (6)
Original Loan Amount (\$)	248,454	248,798	188,800	221,908	50,250	504,400
Loan Amount at Default (\$)	243,996	185,311	219,018	48,386	499,861	
Note Date	3/5/2005	5/23/2005	9/1/2005	19mos, 9 days	6/1/2003	12/1/2006
Default Date	8/13/2007	7/1/2007	25mos, 3 days	3/1/2005	3/1/2010	
Original Note Rate	7.32%	7.32%	7.13%	2.59%	4.75%	10.65%
Credit Score	674.37	668.84	670.00	68.64	576.00	761.00
LTV	71.19	72.02	80.00	22.44	26.70	95.00
Adjustable Rate	0.60	0.63				
Owner Occupied	0.82	0.82				
Purchases	0.41	0.41				
Single Family	0.71	0.70				
Full Documentation	0.31	0.31				

Panel B: summary of loan outcomes						
	BlackBox Proportion (1)	Sample Proportion (2)	Average Months to Completion (3)	SD of Months to Completion (4)	Average NPV (5)	SD of NPV (6)
Defaulted	0.782					
Self Cure		0.263	7.07	5.09	184,385	202,589
Paid-In-Full		0.396	2.37	1.87	153,749	179,134
Modified		0.071	8.31	9.03	251,460	231,944
Foreclosed		0.240	24.51	19.38	161,797	179,350
Liquidation		0.002	12.03	13.01	122,774	149,536
Bankruptcy		0.028			45,737	112,863
Repurchased		0.000			120,481	155,593
					208,398	211,931

Panel C: summary of loan characteristics by outcome						
	Self Cure (1)	Paid-In-Full (2)	Modified (3)	Foreclosed (4)	Liquidated (5)	
Average Original Loan Amount (\$)	220,250	263,701	272,545	255,020	116,405	
Average Loan Amount at Default (\$)	217,057	253,211	273,511	255,714	114,245	
Average Note Date	5/4/2005	11/17/2004	2/6/2006	1/22/2006	1/28/2009	
Average Default Date	6/18/2007	2/9/2007	10/23/2007	3/5/2008	1/24/2009	
Average Original Note Rate	7.41%	7.05%	7.24%	7.66%	8.87%	
Average Credit Score	656.77	679.32	666.36	665.91	691.10	
Average LTV	71.52	69.94	75.14	75.17	39.03	
Proportion, Adjustable Rate	0.58	0.66	0.62	0.67	0.57	
Proportion, Owner Occupied	0.83	0.84	0.88	0.77	0.76	
Proportion, Purchases	0.39	0.39	0.40	0.46	0.44	
Proportion, Single Family	0.71	0.71	0.71	0.68	0.61	
Proportion, Full Documentation	0.38	0.27	0.37	0.29	0.19	

Table 2
First stage.
This table reports regression results relating to the first-stage estimates obtained by regressing a measure of servicer financial constraints, the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* in columns (1) and (2), and the *Credit Default Swap Spread* in columns (3) and (4), on the instrument, the *1-Year Servicer Portfolio Housing Price Return* all measured at the time of borrower first delinquency, and a vector of controls. Columns (1) and (3) represent Eq. (5) at a loan level and column (1) is the specification used as the first stage in the two-stage least squares setup that underlies all of the instrumental variable results. Columns (2) and (4) perform the same regressions at the servicer level, restricting attention to servicer-months that also appear in columns (1) and (3). First-stage coefficient estimates from the servicer level are smaller than at the loan level, implying that the use of the loan-level specification results in a more conservative IV estimate. The instrument utilized for the loan-level specifications is the housing price variation of the servicer's portfolio for geographies outside of the CBSA in which the loan resides, while the servicer-level specifications utilize no geographic holdout. Servicer-level *1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count* and fixed effects are included in all specifications. Additionally, the servicer-level specifications have year fixed effects while the loan-level specifications include the *1-Year Zip Housing Price Return* for the zip code where the loan resides, as well as a vector of loan and borrower characteristic controls and an interacted zip-year-credit category fixed effect. Values of both the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* measure as well as the *Credit Default Swap Spread* have been standardized for the sample utilized in each regression. t-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer level. Columns (1) and (3) are additionally clustered at the zip level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	1-Year Change in Advances as a Percentage of Loan Portfolio Balance		Credit Default Swap Spread	
	Loan (1)	Servicer (2)	Loan (3)	Servicer (4)
1-Year Servicer Portfolio Housing Price Return (Excluding Loan CBSA)	-8.213*** (-5.00)		-11.01*** (-5.48)	
1-Year Servicer Portfolio Housing Price Return (No Geo Holdout)		-3.385*** (-4.95)		-9.131*** (-5.54)
1-Year Zip Housing Price Return	Yes	—	Yes	—
1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count	Yes	Yes	Yes	Yes
Loan/Borrower Characteristic Controls	Yes	—	Yes	—
Servicer FE	Yes	Yes	Yes	Yes
Zip-Year-CreditCat FE	Yes	—	Yes	—
Year FE	—	Yes	—	Yes
Number of observations	6,868,451	9348	1,102,186	494
Adj. R ²	0.432	0.184	0.561	0.580
Incremental Adj. R ²	0.026	0.009	0.035	0.051

5.2 Table 2: First-stage regressions

Read:

- The instrument is the one-year servicer portfolio housing price return excluding the focal loan's CBSA.
- At the loan level, the coefficient is -8.213 , meaning that worse housing price performance elsewhere in the servicer's portfolio predicts higher advance growth.

- At the servicer level, the coefficient is -3.385 .
- Using CDS spreads as an alternative financial-constraint measure also yields negative first-stage coefficients.

Economic message: Housing price declines in a servicer's other markets generate plausibly exogenous servicer-level cash needs. This supports the IV design.

Table 2
First stage.
This table reports regression results relating to the first-stage estimates obtained by regressing a measure of servicer financial constraints, the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* in columns (1) and (2), and the *Credit Default Swap Spread* in columns (3) and (4), on the instrument, the *1-Year Servicer Portfolio Housing Price Return* all measured at the time of borrower first delinquency, and a vector of controls. Columns (1) and (3) represent Eq. (5) at a loan level and column (1) is the specification used as the first stage in the two-stage least squares setup that underlies all of the instrumental variable results. Columns (2) and (4) perform the same regressions at the servicer level, restricting attention to servicer-months that also appear in columns (1) and (3). First-stage coefficient estimates from the servicer level are smaller than at the loan level, implying that the use of the loan-level specification results in a more conservative IV estimate. The instrument utilized for the loan-level specifications is the housing price variation of the servicer's portfolio for geographies outside of the CBSA in which the loan resides, while the servicer-level specifications utilize no geographic holdout. Servicer-level *1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count* and fixed effects are included in all specifications. Additionally, the servicer-level specifications have year fixed effects while the loan-level specifications include the *1-Year Zip Housing Price Return* for the zip code where the loan resides, as well as a vector of loan and borrower characteristic controls and an interacted zip-year-credit category fixed effect. Values of both the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* measure as well as the *Credit Default Swap Spread* have been standardized for the sample utilized in each regression. *t*-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer level. Columns (1) and (3) are additionally clustered at the zip level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	1-Year Change in Advances as a Percentage of Loan Portfolio Balance		Credit Default Swap Spread	
	Loan (1)	Servicer (2)	Loan (3)	Servicer (4)
<i>1-Year Servicer Portfolio Housing Price Return (Excluding Loan CBSA)</i>	-8.213^{***} (-5.00)		-11.01^{***} (-5.48)	
<i>1-Year Servicer Portfolio Housing Price Return (No Geo Holdout)</i>		-3.385^{***} (-4.95)		-9.131^{***} (-5.54)
<i>1-Year Zip Housing Price Return</i>	Yes	—	Yes	—
<i>1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count</i>	Yes	Yes	Yes	Yes
Loan/Borrower Characteristic Controls	Yes	—	Yes	—
Servicer FE	Yes	Yes	Yes	Yes
Zip-Year-CreditCat FE	Yes	—	Yes	—
Year FE	—	Yes	—	Yes
Number of observations	6,868,451	9348	1,102,186	494
Adj. R^2	0.432	0.184	0.561	0.580
Incremental Adj. R^2	0.026	0.009	0.035	0.051

5.3 Table 3: Default intervention

Read:

- The main IV estimate for default intervention is 0.140, with a strong *t*-statistic of 5.07.
- The foreclosure coefficient is 0.0906, meaning constrained servicers are more likely to foreclose after the first delinquency episode.
- The modification coefficient is 0.0526, showing that constrained servicers also modify more.
- Paid-in-full and self-cure effects are negative, especially paid-in-full at -0.0959 .

Economic message: Financial constraints push servicers away from waiting for borrower self-resolution and toward active loss-mitigation actions that end the advance obligation.

5.4 Table 4: Ultimate outcomes

Read:

- The IV coefficient for ultimately foreclosed is 0.117.
- The coefficient for ultimately modified is 0.0600.
- The coefficient for ultimately paid-in-full is -0.0953 .
- The results persist when ultimate rather than first-episode outcomes are used.

Table 3

Default intervention.

This table reports results related to the estimation of a linear probability model regressing a series of loan outcome binary indicators (calculated relative to the outcome of the borrower's first episode of delinquency) on an instrumented *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* measured at the time of borrower first delinquency, as well as a vector of controls and fixed effects. Column (1) additionally reports an instrumented version of column (3), while column (2) reports the commensurate reduced form regression (*Default Intervention* indicator directly on the instrument). Column (4) duplicates column (3), but instead utilizing cumulative levels rather than changes in advances. A default intervention is defined as either a foreclosure (column (5)), a modification (column (6)) or a liquidation. The coefficient estimate in column (3) is predominantly relative to the counterfactuals of *Paid-In-Full* and *Self Cure* that are utilized in columns (7) and (8) respectively. Values of the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* and *Advances as a Percentage of Loan Portfolio Balance* measures have been standardized for the sample utilized in each regression. *t*-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer and zip levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Default Intervention				Foreclosure		Modification		Paid-In-Full		Self Cure	
	OLS (1)	Reduced Form (2)	IV (3)	IV (4)	IV (5)	IV (6)	IV (7)	IV (8)	IV (9)	IV (10)	IV (11)	IV (12)
<i>1-Year Change in Advances as a Percentage of Loan Portfolio Balance</i>	0.0113*** (2.88)		0.140*** (5.07)		0.0906*** (4.51)	0.0526* (1.75)			-0.0959*** (-2.94)		-0.0357 (-1.54)	
<i>Advances as a Percentage of Loan Portfolio Balance</i>				0.220*** (4.16)								
<i>1-Year Servicer Portfolio Housing Price Return (Excluding Loan CISA)</i>		-1.154*** (-7.91)										
<i>1-Year Zip Housing Price Return</i>	-0.721*** (-16.28)	-0.291*** (-7.17)	-0.473*** (-10.40)	-0.484*** (-9.61)	-0.368*** (-7.30)	-0.103** (-1.98)	0.379*** (4.85)	0.124** (1.97)				
<i>1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count</i>	-0.000397 (-0.44)	0.000153 (0.17)	0.00496 (0.85)	0.00332 (1.61)	0.00509 (1.06)	-0.000179 (-0.11)	-0.00708 (-1.55)	0.00125 (0.55)				
Loan/Borrower Characteristic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Servicer FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip-Year-CreditCat FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451
Adj. R ²	0.260	0.262							0.001			
Dependent Variable Average	0.3135	0.3135	0.3135	0.3135	0.2404	0.0713	0.3961	0.2618				

Economic message: Servicer financial constraints do not merely change short-run classifications. They affect eventual loan resolution outcomes.

Table 4

Ultimate outcomes.

This table reports results related to the estimation of a linear probability model regressing a series of loan outcome binary indicators on an instrumented *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* measured at the time of borrower first delinquency, as well as a vector of controls and fixed effects. Column (1) additionally reports an instrumented version of column (3), while column (2) reports the commensurate reduced form regression (*Ultimately Foreclosed* indicator directly on the instrument). Column (4) duplicates column (3), but instead utilizing cumulative levels rather than changes in advances. Outcome indicators utilized in this table are calculated in reference to ultimate outcomes on the loan. That is, regardless of the initial result of the borrower's first episode of delinquency, was this loan ever placed into foreclosure (column (3)), modified (column (5)), or paid off in full (column (6)). Values of the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* and *Advances as a Percentage of Loan Portfolio Balance* measures have been standardized for the sample utilized in each regression. *t*-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer and zip levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Ultimately Foreclosed				Ultimately Modified		Ultimately Paid-In-Full	
	OLS (1)	Reduced Form (2)	IV (3)	IV (4)	IV (5)	IV (6)	IV (7)	IV (8)
<i>1-Year Change in Advances as a Percentage of Loan Portfolio Balance</i>	0.0156*** (3.59)		0.117*** (5.45)		0.0600* (1.78)	-0.0953*** (-2.94)		
<i>Advances as a Percentage of Loan Portfolio Balance</i>				0.184*** (4.43)				
<i>1-Year Servicer Portfolio Housing Price Return (Excluding Loan CISA)</i>		-0.964*** (-7.53)						
<i>1-Year Zip Housing Price Return</i>	-0.595*** (-7.27)	-0.248*** (-3.55)	-0.401*** (-5.50)	-0.409*** (-5.62)	-0.193*** (-3.75)	0.380*** (4.86)		
<i>1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count</i>	0.00137 (1.21)	0.00158 (1.33)	0.00560 (1.04)	0.00423* (1.92)	-0.000716 (-0.46)	-0.00708 (-1.55)		
Loan/Borrower Characteristic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Servicer FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip-Year-CreditCat FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451
Adj. R ²	0.262	0.263	0.263	0.263	0.003	0.003		
Dependent Variable Average	0.3986	0.3986	0.3986	0.3986	0.1646	0.3963		

Table 4

Ultimate outcomes.

This table reports results related to the estimation of a linear probability model regressing a series of loan outcome binary indicators on an instrumented *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* measured at the time of borrower first delinquency, as well as a vector of controls and fixed effects. Column (1) additionally reports an instrumented version of column (3), while column (2) reports the commensurate reduced form regression (*Ultimately Foreclosed* indicator directly on the instrument). Column (4) duplicates column (3), but instead utilizing cumulative levels rather than changes in advances. Outcome indicators utilized in this table are calculated in reference to ultimate outcomes on the loan. That is, regardless of the initial result of the borrower's first episode of delinquency, was this loan ever placed into foreclosure (column (3)), modified (column (5)), or paid off in full (column (6)). Values of the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* and *Advances as a Percentage of Loan Portfolio Balance* measures have been standardized for the sample utilized in each regression. *t*-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer and zip levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Ultimately Foreclosed				Ultimately Modified		Ultimately Paid-In-Full	
	OLS (1)	Reduced Form (2)	IV (3)	IV (4)	IV (5)	IV (6)	IV (7)	IV (8)
<i>1-Year Change in Advances as a Percentage of Loan Portfolio Balance</i>	0.0156*** (3.59)		0.117*** (5.45)		0.0600* (1.78)	-0.0953*** (-2.94)		
<i>Advances as a Percentage of Loan Portfolio Balance</i>				0.184*** (4.43)				
<i>1-Year Servicer Portfolio Housing Price Return (Excluding Loan CISA)</i>		-0.964*** (-7.53)						
<i>1-Year Zip Housing Price Return</i>	-0.595*** (-7.27)	-0.248*** (-3.55)	-0.401*** (-5.50)	-0.409*** (-5.62)	-0.193*** (-3.75)	0.380*** (4.86)		
<i>1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count</i>	0.00137 (1.21)	0.00158 (1.33)	0.00560 (1.04)	0.00423* (1.92)	-0.000716 (-0.46)	-0.00708 (-1.55)		
Loan/Borrower Characteristic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Servicer FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Zip-Year-CreditCat FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451
Adj. R ²	0.262	0.263	0.263	0.263	0.003	0.003		
Dependent Variable Average	0.3986	0.3986	0.3986	0.3986	0.1646	0.3963		

Table 5

Financial constraints exacerbate agency frictions.

This table reports results demonstrating that servicers act differently according to the severity of their agency frictions. Column (1) replicates the instrumental variables specification presented in column (3) of Table 3. Columns (2) and (3) update column (1) by including a *High-Quality Loan Indicator* as well as an interaction term between that indicator and the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance*. These instrumental variable specifications utilize *1-Year Servicer Portfolio Housing Price Return (Excluding Loan CISA)* as well as the interaction of that variable with the *High-Quality Loan Indicator* to instrument for the advances variable and its interaction. Column (4) replicates the specification in column (3) in the setting of column (3) of Table 4. Columns (3) and (4) drop from the *Loan/Borrower Characteristic Controls* set both the origination FICO and origination LTV and also reduce the interacted zip code, delinquency year, credit category fixed effect dimension by removing credit category. The value of the *F*-statistic testing whether the sum of the coefficients for the advances term and the interaction term is equal to zero, along with the corresponding *p*-value is also reported. Values of the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* measure have been standardized for the sample utilized in each regression. *t*-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer and zip levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Default Intervention				Ultimately Foreclosed	
	IV (1)	IV (2)	IV (3)	IV (4)	IV (5)	IV (6)
<i>1-Year Change in Advances as a Percentage of Loan Portfolio Balance</i>	0.140*** (5.07)	0.146*** (5.16)	0.138*** (4.92)	0.114*** (5.28)		
<i>High-Quality Loan Indicator</i>		-0.057*** (-3.81)	-0.112*** (-9.98)	-0.158*** (-11.27)		
<i>1-Year Change in Advances as a Percentage of Loan Portfolio Balance</i>		-0.111*** (-4.34)	-0.0943*** (-3.45)	-0.0964*** (-3.58)		
<i>× High-Quality Loan Indicator</i>						
<i>1-Year Zip Housing Price Return</i>	-0.473*** (-10.40)	-0.473*** (-10.65)	-0.468*** (-9.97)	-0.401*** (-5.28)		
<i>1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count</i>	0.00496 (0.85)	0.00474 (0.83)	0.00470 (0.84)	0.00538 (1.06)		
Loan/Borrower Characteristic Controls	Yes	Yes	Yes	Yes	Yes	Yes
Loan/Borrower Characteristic Controls (Adj)	Yes	Yes	Yes	Yes	Yes	Yes
Servicer FE	Yes	Yes	Yes	Yes	Yes	Yes
Zip-Year-CreditCat FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451
Adj. R ²					0.003	
<i>F</i> -Stat: Advances + Interaction = 0		(0.94)	(1.40)	(0.40)		
<i>p</i> -Value		0.33	0.24	0.53		

5.5 Table 5: Agency frictions and high-quality loans

Read:

- Column 1 replicates the baseline IV intervention result from Table 3.
- High-quality loans have lower intervention probabilities.
- The interaction between financial constrainedness and the high-quality indicator is negative and significant.
- For default intervention, the interaction is around -0.111 to -0.0943 .
- For ultimately foreclosed, the interaction is -0.0964 .

Economic message: Financial constraints matter less when agency frictions are lower. This supports the interpretation that constrained servicers exploit discretion when investor monitoring is more difficult.

5.6 Table 6: Investor value

Read:

- The main IV estimate implies that a one-standard-deviation increase in financial constrainedness reduces investor NPV by about \$21,663.
- The high-quality interaction is positive, about \$31,648, indicating that the NPV damage of financial constraints is largely absent for high-quality loans.
- The high-quality indicator itself is positive, consistent with higher investor value for better-quality borrowers.

Economic message: Constrained servicers destroy investor value. The result is not merely about faster intervention; it is about inefficient intervention.

Table 6
Investor value.
This table reports results demonstrating the causal impact of a servicer's financial constraints, as measured at the time of borrower first delinquency by the 1-Year Change in Advances as a Percentage of Loan Portfolio Balance, on the NPV for the investor. Column (1) reports the coefficient estimate for an uninstrumented OLS specification, column (2) displays the reduced form specification wherein investor NPV is regressed directly on the instrument 1-Year Servicer Portfolio Housing Price Return (Excluding Loan CREA), and column (3) utilizes an instrumental variables framework to report the causal estimate. Column (4) duplicates column (3), but instead utilizing cumulative levels rather than changes in advances. Column (5) replicates the setting in Table 5 including an indicator for whether the focal loan was high-quality as well as an interaction of that indicator with the advances measure of financial constraints. Advances are instrumented as in column (3), whereas the interaction term is instrumented with the interaction of the indicator and that same instrument. The standard controls and fixed effects are utilized throughout. Values of the 1-Year Change in Advances as a Percentage of Loan Portfolio Balance and Advances as a Percentage of Loan Portfolio Balance measures have been standardized for the sample utilized in each regression. t-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer and zip levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Investor NPV				
	OLS (1)	Reduced Form (2)	IV (3)	IV (4)	IV (5)
1-Year Change in Advances as a Percentage of Loan Portfolio Balance	-1,906.0** (-2.03)		-21,663.0*** (-3.62)		-22,604*** (-3.67)
Advances as a Percentage of Loan Portfolio Balance				-33,993.8*** (-3.27)	
High-Quality Loan Indicator					18,485*** (5.87)
1-Year Change in Advances as a Percentage of Loan Portfolio Balance × High-Quality Loan Indicator					31,648*** (8.31)
1-Year Servicer Portfolio Housing Price Return (Excluding Loan CREA)		177,918.3*** (4.93)			
1-Year Zip Housing Price Return	99,470.9*** (6.33)	33,574.3* (1.67)	61,658.2*** (3.46)	63,225.4*** (3.50)	60,648*** (3.37)
1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count	-108.3 (-0.33)	-186.4 (-0.57)	-927.3 (-0.90)	-675.5 (-1.59)	-857.0 (-0.87)
Loan/Borrower Characteristic Controls	Yes	Yes	Yes	Yes	Yes
Loan/Borrower Characteristic Controls (Adj)	-	-	-	-	Yes
Servicer FE	Yes	Yes	Yes	Yes	Yes
Zip-Year-CreditCat FE	Yes	Yes	Yes	Yes	Yes
Zip-Year FE	-	-	-	-	Yes
Number of observations	6,868,451	6,868,451	6,868,451	6,868,451	6,868,451
Adj. R ²	0.830	0.830	0.670	0.671	0.674
Dependent Variable Average	185,130	185,130	185,130	185,130	185,130
F-Stat: Advances + Interaction = 0					2.39
p-Value					0.12

Table 7
Performance of foreclosures and modifications.
This table reports results relating to the performance and characteristics of foreclosures and modifications performed on loans following their first episode of delinquency. Column (1) reports, conditional on a loan being foreclosed on during its first episode of delinquency, the results from regressing the Foreclosure Liquidation Loss Severity on an instrumented measure of servicer financial constraints, the 1-Year Change in Advances as a Percentage of Loan Portfolio Balance measured at the time of borrower first delinquency, and the standard controls and fixed effects. Columns (2) and (3) report the results from regressing the percentage point reduction in interest rate associated with a modification on an instrumented measure of servicer financial constraints, the 1-Year Change in Advances as a Percentage of Loan Portfolio Balance measured at the time of borrower first delinquency, and the standard controls and fixed effects. Columns (4) and (5) regress indicators of whether the loan again became delinquent three and twelve months respectively, after the modification brought the loan current on the measure of servicer financial constraints and the standard controls and fixed effects. Columns (2)-(5) condition on a loan being modified during its first episode of delinquency. Columns (2) and (3) further condition on the loan being a fixed rate loan both prior and subsequent to the modification, and column (3) on the modification performed containing some interest rate reduction component. Values of the 1-Year Change in Advances as a Percentage of Loan Portfolio Balance measure have been standardized for the sample utilized in each regression. t-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer and zip levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Foreclosure Liquidation Loss Severity		Modification Interest Rate Reduction		3mo Redefault	12mo Redefault
	FC Only IV (1)	FC Only IV (2)	Fixed Rate Mods IV (3)	FR Int Rt Red. IV (4)	Mod Only IV (5)	Mod Only IV (6)
1-Year Change in Advances as a Percentage of Loan Portfolio Balance	0.0400** (2.49)	1.689*** (2.98)	0.727* (1.86)		0.0704* (1.97)	0.0618* (1.82)
1-Year Zip Housing Price Return	-0.108*** (-3.61)	-0.829 (-1.39)	0.120 (0.39)		-0.0794 (-1.34)	-0.0855 (-1.11)
1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count	5.32e-06 (0.00)	-0.211 (-0.70)	-0.419 (-0.46)		0.0369** (2.24)	0.0303* (1.67)
Loan/Borrower Characteristic Controls	Yes	Yes	Yes	Yes	Yes	Yes
Servicer FE	Yes	Yes	Yes	Yes	Yes	Yes
Zip-Year-CreditCat FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	1,353,649	110,219	33,325		440,921	440,921
Adj. R ²	0.050		0.143			0.004
Dependent Variable Average	68.03%	1.16%	3.05%		0.3498	0.5426

5.7 Table 7: Performance of foreclosures and modifications

Read:

- Conditional on foreclosure, constrained servicers' foreclosures have higher liquidation loss severity.
- Conditional on modification, constrained servicers provide larger interest rate reductions.
- Modifications by constrained servicers have higher 3-month and 12-month redefault rates.
- These patterns indicate that constrained servicers produce lower-quality interventions.

Economic message: Financial constraints affect not just whether servicers intervene but how effectively they intervene.

5.8 Table 8 and Table 9: Conditional investor value and borrower-quality checks

Read:

- Table 8 shows that conditional on foreclosure, constrainedness reduces investor NPV by about \$10,727.
- Conditional on modification, constrainedness reduces investor NPV by about \$6,382.

- Liquidation-only results are not statistically meaningful.
- Table 9 shows that servicer financial constrainedness is not strongly related to measures of unobservable borrower quality at origination.
- The borrower-quality proxies include delinquency probabilities, months to first delinquency, and early payment habits.

Economic message: Table 8 strengthens the value-destruction result, while Table 9 reduces concern that the instrument simply sorts risky borrowers to constrained servicers.

Table 8

Conditional investor value.

This table reports instrumental variable regression results similar to that of column (3) in Table 6 with further conditioning based on the servicer action taken after the first episode of borrower delinquency. Columns (1)-(3) condition on foreclosure, modification, and liquidation (generally a charge-off or a short sale) respectively. Values of the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* measure have been standardized for the sample utilized in each regression. t-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer and zip levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Investor NPV		
	FC Only IV (1)	Mod Only IV (2)	Liq Only IV (3)
<i>1-Year Change in Advances as a Percentage of Loan Portfolio Balance</i>	-10,727.0*** (-3.00)	-6,382.1** (-2.03)	-63.38 (-0.02)
<i>1-Year Zip Housing Price Return</i>	12,330.9* (1.68)	-31,785.5* (-1.93)	6,292.2 (0.32)
<i>1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count</i>	85.28 (0.21)	0.419 (0.00)	11,692.5 (0.30)
Loan/Borrower Characteristic Controls	Yes	Yes	Yes
Servicer FE	Yes	Yes	Yes
Zip-Year-CreditCat FE	Yes	Yes	Yes
Number of observations	1,592,194	440,921	7306
Adj. R ²	0.576	0.588	0.655
Dependent Variable Average	123,560	168,093	33,610

Table 8: Conditional investor value

Table 9

Unobservable borrower quality.

This table regresses various measures of unobservable borrower quality on an instrumented measure of servicer financial constraints, the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance*, measured at the time of loan origination. Columns (1) and (2) utilize the full sample of loans in the Blackbox dataset (unconditional on delinquency) and regress an indicator of delinquency over a 1-year and 5-year (post-origination) window, respectively, on the servicer's financial constraints at the time of loan origination. Column (3) regresses the number of months between loan origination and the borrower's first delinquency on the servicer's financial constraints at the time of loan origination. Columns (4)-(6) utilize a measure of unobservable borrower quality related to the frequency with which the borrower makes their monthly payment at least a day before it is due. This measure has been established in the literature as highly correlated with future loan delinquency and default outcomes and is, crucially, unobservable at the time of loan origination. Values of the *1-Year Change in Advances as a Percentage of Loan Portfolio Balance* measure have been standardized for the sample utilized in each regression. t-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer and zip levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	DQ Prob 1-Year IV (1)	DQ Prob 5-Year IV (2)	Months To First DQ IV (3)	Early Payments, Mos 1-6 IV (4)	Early Payments, Mos 4-6 IV (5)	Early Payments, Mos 4-9 IV (6)
<i>1-Year Change in Advances as a Percentage of Loan Portfolio Balance At Origination</i>	0.0309 (0.55)	0.0375 (0.77)	-3.145 (-1.45)	-4.371 (-1.03)	-0.244 (-0.87)	-0.928 (-0.97)
<i>1-Year Zip Housing Price Return</i>	-0.0358 (-0.65)	-0.0572 (-1.16)	4.807* (1.84)	-0.169 (-0.06)	0.190 (0.70)	0.501 (0.75)
<i>1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count, At Origination</i>	2.65e-04 (0.57)	-2.02e-04 (-0.35)	-0.0252 (-0.68)	-6.950 (-1.09)	-0.0564 (-0.83)	-0.203 (-0.88)
Loan/Borrower Characteristic Controls	Yes	Yes	Yes	Yes	Yes	Yes
Servicer FE	Yes	Yes	Yes	Yes	Yes	Yes
Zip-Orig. Year-CreditCat FE	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	8,889,284	8,889,284	6,363,798	2478	2,514,099	2,227,485
Adj. R ²	0.012	0.016	0.013			
Dependent Variable Average	0.25	0.74	25.47	1.87	0.93	1.84

Table 9: Unobservable borrower quality

5.9 Table 10: Capacity constraints

Read:

- Column 1 replicates the baseline intervention result.
- Column 2 omits the capacity control and obtains a similar coefficient.
- When servicer capacity constraints are low, the financial-constraint coefficient is 0.245.
- When servicer capacity constraints are high, the coefficient is only 0.133.

Economic message: Balance-sheet constraints are strongest when the servicer still has operational capacity to act. When capacity is strained, even financially constrained servicers cannot respond as aggressively.

Table 10

Capacity constraints.

This table reports, in column (1), a regression identical to column (3) of Table 3. Column (2) mimics column (1), but with the control for capacity constraints omitted entirely. Columns (3) and (4) report results from specifications identical to those of column (1), but with additional sample conditioning. Column (3) demonstrates the results conditioning on loans experiencing their first delinquency in months where the servicer has lower capacity constraints relative to the median measured capacity constraint for that servicer. Column (4), similarly, reports results in the sample where the servicer has above-median capacity constraints. *t*-statistics in parentheses are heteroskedasticity-robust and clustered at the servicer and zip levels. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Default Intervention		Default Intervention	
	IV (1)	IV (2)	Low Capacity Constraint IV (3)	High Capacity Constraint IV (4)
<i>1-Year Change in Advances as a Percentage of Loan Portfolio Balance</i>	0.140*** (5.07)	0.140*** (5.16)	0.245*** (2.76)	0.133*** (4.54)
<i>1-Year Change in Delinquency Count As % of Servicer Portfolio Loan Count</i>	0.00496 (0.85)		-0.296 (-0.92)	0.00249 (0.45)
<i>1-Year Zip Housing Price Return</i>	-0.473*** (-10.40)	-0.474*** (-10.45)	-0.517*** (-6.86)	-0.525*** (-9.57)
Loan/Borrower Characteristic Controls	Yes	Yes	Yes	Yes
Servicer FE	Yes	Yes	Yes	Yes
Zip-Year-CreditCat FE	Yes	Yes	Yes	Yes
Number of observations	6,868,451	6,868,451	3,349,398	3,469,802
Adj. R^2				

6 Short summary

- Mortgage servicers must advance missed payments to MBS investors when borrowers are delinquent.
- These advances create a balance-sheet burden that can distort servicer incentives.
- The paper measures servicer financial constrainedness with changes in advances relative to portfolio balance.
- It instruments constrainedness with housing price returns in the servicer's other markets, excluding the focal loan's CBSA.
- Financially constrained servicers intervene more aggressively after the first delinquency episode.
- They foreclose more, modify more, and allow fewer paid-in-full or self-cure outcomes.
- These interventions reduce investor NPV by more than \$20,000 per defaulted loan in the main IV estimate.
- The effect is weaker for high-quality loans, where agency frictions are lower.
- Conditional on intervention, constrained servicers also perform worse: foreclosures have higher loss severity and modifications redefault more.
- Borrower-quality tests reduce concern that the results are driven by unobserved borrower sorting.
- Capacity constraints dampen the responsiveness of servicers to financial constraints.
- The paper shows that mortgage servicing institutions can transmit financial stress into real foreclosure and investor-value losses.

7 Conclusion

7.1 Core conclusion

Aiello shows that financially constrained mortgage servicers destroyed substantial investor value during the financial crisis. Servicers that faced larger advance burdens were more likely to intervene in delinquent loans through foreclosure and modification, and those interventions were often worse for investors.

7.2 Economic intuition

The core intuition is that servicing advances create a financing obligation. A patient servicer can advance delinquent payments and allow a borrower time to self-cure or pay off. A financially constrained servicer

has stronger incentives to end the advance obligation quickly, even if premature foreclosure or modification sacrifices long-run investor value.

7.3 Takeaway

Mortgage servicing is a key agency link in securitized credit markets. The paper demonstrates that servicer financial constraints and operational capacity can shape foreclosure decisions, modification quality, and investor losses. Designing mortgage contracts and crisis policies therefore requires attention not only to borrowers and investors, but also to the balance sheets and incentives of servicers.